Borel and residue-kernel atoms in the Higman–Sims catalogue

Private L091/L096 finite component

The fixed source is Dimitri Leemans's 23-page Higman–Sims supplement, dated 6 October 2003, retained with the L096 materials. Diagram numbers refer to that edition. We acquire the reviewed 86-family classification, its literal source-coordinate bijection, and the computed Borel and residue-kernel generators. This note supplies a first structural batch, without another subgroup or geometry census.

Theorem 1. *The 350 occurrences in the literal source-join ledger have the structures listed below. They comprise 55 Borel groups and 295 residue kernels. Among these occurrences, the only changes to the printed group labels are the Borels of diagrams 3.15 and 3.16: both are* $C_3^2 : Q_8$*, in place of the printed* $2^4 : A_4$*.*

Structure	Borels	Kernels
A_4	1	0
A_5	2	1
1	0	204
C_2	1	26
C_3	3	5
C_4	4	5
C_4^3	0	1
$C_5 : C_4$	3	5
C_6	1	1
D_{12}	3	0
D_8	9	0
C_2^2	10	12
C_2^3	0	11
C_2^4	0	12
$C_3^2 : Q_8$	5	1
$L_3(4)$	1	1
Q_8	7	8
S_4	2	0
S_6	3	2
Total	55	295

Here D_n denotes the dihedral group of order n, and powers of C_m denote direct powers, not an unspecified group of the same order. The remaining 31 Borels and 17 kernels are listed individually in `remaining.json`. No parabolic structure or quotient structure is asserted merely from one of these kernel identifications.

The actual subgroup interface. For the acquired right-coset geometry let G_i be its standard parabolics. The Borel is $B = \bigcap_i G_i$. The residue kernel at type i is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The reviewed local-data component proves this formula and records the actual generators of these groups on the same 100 points. We use those generators literally. The source-coordinate permutation specifies which model type belongs to each printed node; equal orders do not choose that association. The source's box denotes the kernel even when it occurs inside parentheses, as in $(\boxed{2} \times A_6) \cdot 2^2$. The dot separating a kernel and quotient is generic extension notation; it is not an additional nonsplitting assertion.

Lemma 1 (Literal generator-edge transport)**.** *Let $M = \langle s_1, \ldots, s_t \rangle$ and $U = \langle u_1, \ldots, u_v \rangle$ be finite permutation groups. Suppose a map $f : M \longrightarrow U$ is a bijection, $f(1) = 1$, and for fixed $h_i \in U$ it satisfies $f(xs_i) = f(x)h_i$ for every $x \in M$ and every i. Then f is an isomorphism carrying s_i to h_i.*

Proof. Every element has a word in the generators and their inverses. Since the groups are finite, inverse powers can also be replaced by positive powers. Induction on a word for y gives $f(xy) = f(x)f(y)$. Thus f is a bijective homomorphism. This is the acquired marked-coordinate transport argument, here applied to the small literal subgroup closures. □

The certificate records the images of the ordered model generators. The reader computes the entire model closure and every right-generator edge, propagating images from the identity. Every repeated encounter must give the same image. It independently computes the actual subgroup closure, checks image membership, and compares the complete element sets bijectively. Thus the generator images are compact data for a full finite certificate, not an unverified isomorphism proposal. The 350 checks use 92,530 edges.

Elementary models. Cyclic and homocyclic groups use disjoint cycles. Their full distinct normal forms are checked; in particular C_4^3 is not inferred from order 64. Dihedral models use rotation and reversal on a polygon. The quaternion model satisfies $a^4 = 1$, $b^2 = a^2$, and $b^{-1}ab = a^{-1}$, with the eight distinct forms $a^i b^j$, $0 \leq i < 4$, $0 \leq j < 2$. For S_n and A_n, $n \leq 6$, the generated element set is compared with all permutations, respectively all even permutations, of the specified points. The order-20 model is the affine group of the five-element field: translations and multiplication by 2 give an actual $C_5 : C_4$ decomposition. The sole non-elementary named atom, $L_3(4)$, is acquired from the accepted L100 projective construction through the reviewed M22 and linear-table registry. Its exact 21-point generators are checked, not replaced by an order-based recognition. The named projective-group interface remains an explicit inherited premise.

The five Borels of order 72. For completeness we specify the relevant affine model directly. On $V = \mathbf{F}_3^2$, use the two coordinate translations together with the right linear actions of

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

They satisfy $A^2 = C^2 = -I$ and $AC = -CA$. Their eight distinct products $A^i C^j$ form Q_8. The linear group fixes the zero vector, whereas the translation group acts regularly. Hence their intersection is trivial, translations are normal, and their product is $V \rtimes Q_8$ of order 72. The reader checks all these assertions on the nine actual points, including the distinct normal forms and nontrivial complement action. It also checks that every nonidentity element of the complement fixes no nonzero vector.

The literal transport certificates identify the Borels of diagrams 3.12, 3.13, 3.14, 3.15 and 3.16 with this model. The corresponding fixed representatives are 24, 21, 23, 25 and 22. The kernel of source type 2 in 3.14 (model type 1 in representative 23) has the same identification. The first three printed Borel labels already say $3^2 : Q_8$. The last two instead say $2^4 : A_4$, which has order 192. Their native pages 6 and 7 have been inspected directly. Thus the correction is supported both by the acquired exact order and by a positive structural certificate. It does not alter the principal geometry classification.

Source checks, validation and limits. The full ledger records the source page, diagram, printed label, source node, model node, actual order and proved model for each occurrence. The source transcription and coordinate theorem are acquired from their separate reviews. A native recheck of pages 7–8 caught an initial extraction mistake: four boxes inside parentheses had been incorrectly treated as absent. The predecessor diagnostic is preserved. Correct extraction agrees with all four printed C_2 kernels; no source correction is asserted there.

One capped GAP invocation proposed the finite maps in 2.26 seconds. Its receipt and all proposals are retained. The final proof reader calls neither GAP nor an isomorphism oracle. Its ordinary and optimized runs must agree. Damaged map images, wrong subgroup orders and generators, an incorrect same-order model, an elementary substitute for C_4^3, malformed metadata and interior-box examples exercise the certificate boundaries. An initial negative control expected a later bijection failure but correctly failed earlier at a generator edge; that harness failure is preserved.

The work acquires prior mathematical classifications and literal models; it does not reprove those classifications. The new model checks, transport application, source extraction and program are by the present component's author. Root assigned the scope and the two known order discrepancies; the atlas author supplied registry pointers. No whole-source, new question, formalization or measurement credit is claimed. Compound groups outside the 350-occurrence ledger, all parabolic structures, other sporadic source clauses, and external rank-six construction attributions remain separate.